



**SYDNEY BOYS HIGH SCHOOL**  
MOORE PARK, SURRY HILLS

**2002**  
**TRIAL HIGHER SCHOOL**  
**CERTIFICATE EXAMINATION**

# Mathematics

## **General Instructions**

- Reading time – 5 minutes.
- Working time – 3 hours
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may not be awarded for messy or badly arranged work.

## **Total Marks - 120 marks**

- All questions are of equal value.

Examiners: *P. Bigelow, P. Parker*

**NOTE:** This is a trial paper only and does not necessarily reflect the content or format of the final Higher School Certificate Examination Paper for this subject.

**Question 1: (12 Marks)****Marks**

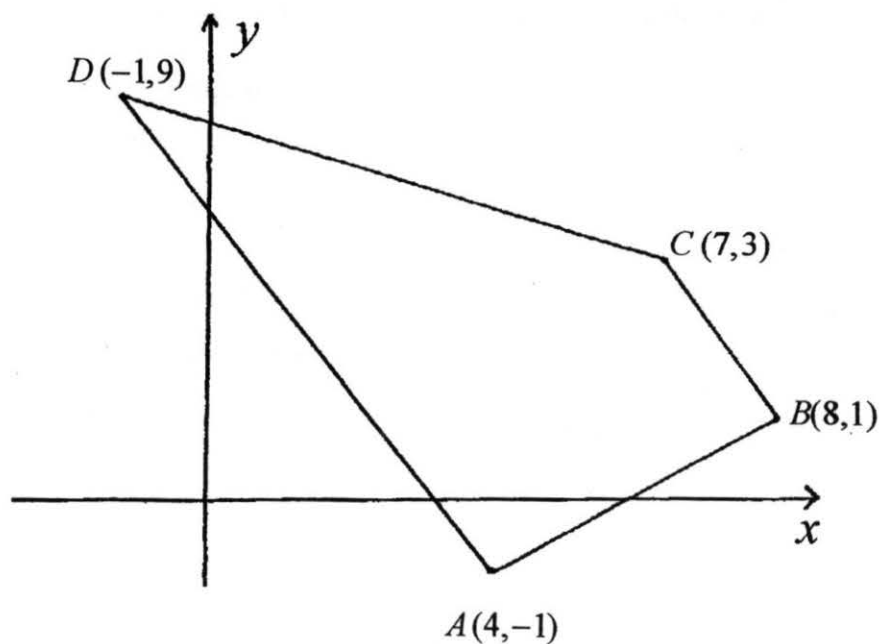
- (a) Solve  $\frac{3}{x} = \frac{4}{5}$  2
- (b) If  $a = 2.673$  and  $b = 1.049$ , evaluate  $\frac{a^2 + b^2}{ab}$  correct to 2 decimal places. 2
- (c) Factorise  $cd - c - dy + y$  2
- (d) The line  $kx - y = 29$  passes through the point  $(4, -1)$ , find the value of  $k$ . 2
- (e) Graph the solution to  $|x + 3| \leq 1$  on a number line. 2
- (f) Find integers  $a$  and  $b$  such that 2

$$\frac{1}{\sqrt{3} + 2} = a\sqrt{3} + b$$

# START A NEW BOOKLET

## Question 2: (12 Marks)

Marks



$A$ ,  $B$ ,  $C$  and  $D$  are the points  $(4, -1)$ ,  $(8, 1)$ ,  $(7, 3)$  and  $(-1, 9)$  respectively.

- |     |                                                      |   |
|-----|------------------------------------------------------|---|
| (a) | Show that the equation of $AC$ is $4x - 3y - 19 = 0$ | 2 |
| (b) | Show $BC \parallel AD$                               | 2 |
| (c) | Show $\angle ACD = 90^\circ$                         | 2 |
| (d) | Show the length of $AC$ is 5 units                   | 2 |
| (e) | Find the perpendicular distance of $B$ from $AC$     | 2 |
| (f) | Find the area of the trapezium $ABCD$ .              | 2 |

# START A NEW BOOKLET

## Question 3: (12 Marks)

Marks

(a) Differentiate

(i)  $(2x-1)^7$  1

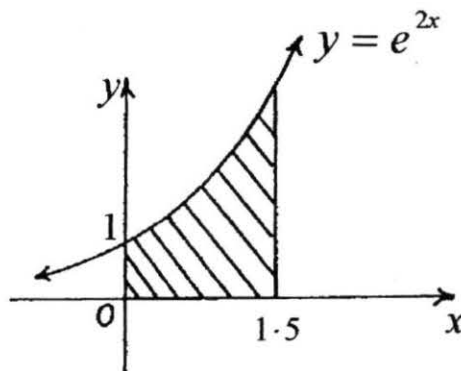
(ii)  $\log_e(4+5x)$  2

(iii)  $x^2 \sin x$  2

(b) Find  $\int \sec^2 3x \, dx$  1

(c) Evaluate  $\int_1^{e^2} \frac{3}{x} \, dx$  3

(d) 3



The diagram above shows the region bounded by the curve  $y = e^{2x}$ , the line  $x = 1.5$  and the coordinate axes.  
Find the EXACT area of this region.

# START A NEW BOOKLET

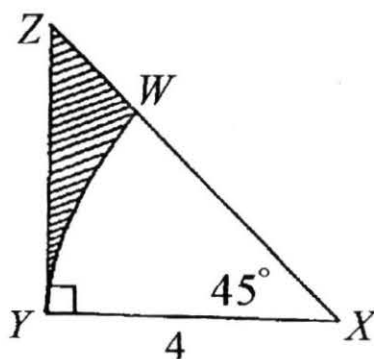
## Question 4: (12 Marks)

Marks

- (a) A parabola has equation  $x^2 - 2x + 25 = 8y$ .
- (i) By completing the square, express this in the general form  $(x - p)^2 = 4a(y - q)$  2
- (ii) State the coordinates of the vertex. 1
- (iii) State the coordinates of the focus. 1
- (iv) State the equation of the directrix. 1
- (b) Find the equation of the line through the intersection of the lines  $x - y = 0$  and  $x + 2y - 6 = 0$  and which is parallel to the line  $3x - 2y + 7 = 0$ . 4

Leave your answer in general form.

- (c) 3



In the diagram above,  $XYZ$  is a right-angled triangle,  $\angle ZXY = 45^\circ$  and  $XY = 4$  units. A circular arc, centre  $X$  and radius  $YX$  cuts the side  $XZ$  at  $W$ .

Find the EXACT area of the shaded region  $WZY$ .

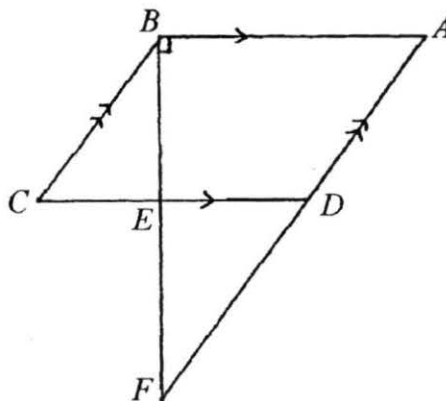
# START A NEW BOOKLET

## Question 5: (12 Marks)

Marks

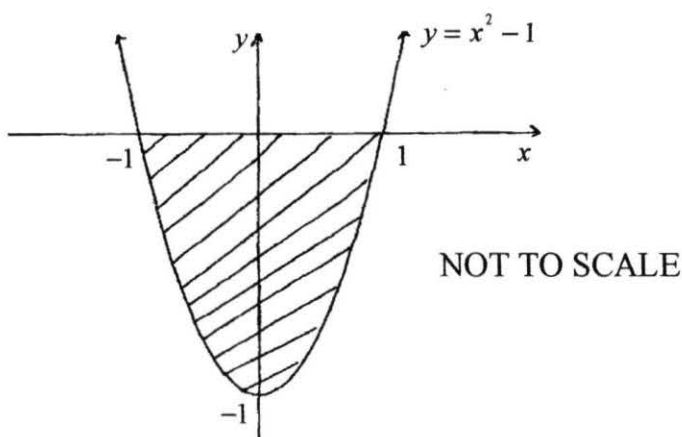
- (a) Find the equation of the normal to  $y = x^2 - 3x + 5$  at the point  $(3, 5)$ .  
Leave your answer in general form. 2

(b)



$ABCD$  is a parallelogram and  $FB \perp AB$

- (i) Prove  $\triangle CBE \parallel \triangle AFB$  3
- (ii) If  $CE = 3$  cm,  $BC = 7$  cm and  $AF = 15$  cm, find  $AB$ . 2
- (c) 3



The area bounded by the curve  $y = x^2 - 1$  and the  $x$ -axis is rotated about the  $x$ -axis.  
Find the volume of the solid of rotation. Leave your answer in EXACT form.

- (d) A continuous curve  $y = f(x)$  has the following properties for the closed interval  $x_1 \leq x \leq x_2$ : 2

$$f(x) > 0, \quad f'(x) < 0, \quad f''(x) > 0$$

Sketch a curve satisfying these conditions.

# START A NEW BOOKLET

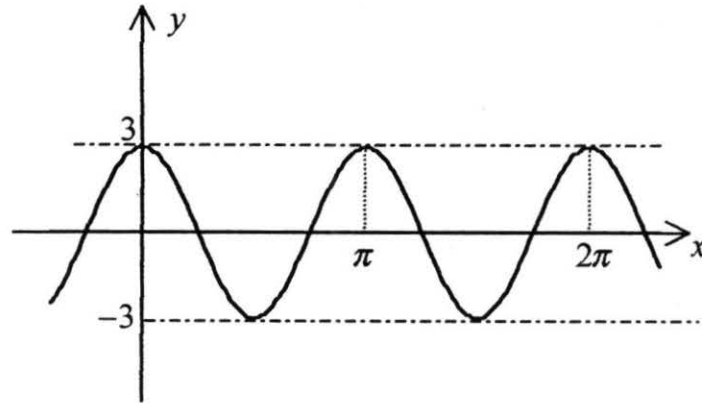
## Question 6: (12 Marks)

Marks

- (a) The curve  $y = x^3 + mx + n$  has a stationary point at  $P(1,5)$ . Find the values of the constants  $m$  and  $n$ .

3

(b)



The graph above represents  $y = a \cos mx$ .

2

Write down the values of  $a$  and  $m$ .

- (c) Express as a single logarithm in simplest form

2

$$\ln 2 + 2 \ln 18 - \frac{3}{2} \ln 36$$

- (d) Simplify  $\sqrt{\frac{1 - \cos^2 \theta}{1 + \tan^2 \theta}}$

2

- (e) The table shows the value of a function  $f(x)$  for five values of  $x$ .

3

|        |    |   |   |    |    |
|--------|----|---|---|----|----|
| $x$    | -1 | 1 | 3 | 5  | 7  |
| $f(x)$ | 5  | 9 | 2 | -1 | -6 |

Use Simpson's rule with these five values to estimate  $\int_{-1}^7 f(x) dx$

## START A NEW BOOKLET

### Question 7: (12 Marks)

Marks

- (a) The value  $\$V$  of a particular make of car can be calculated using the equation

$$V = 50000e^{-0.25t}, \text{ where } t \text{ years is the age of the car}$$

- |       |                                                                                                    |   |
|-------|----------------------------------------------------------------------------------------------------|---|
| (i)   | State the value of the car when it is new                                                          | 1 |
| (ii)  | What is the value of the car after 3 years? (to the nearest \$100).                                | 2 |
| (iii) | How long will it take for the car to be worth \$15 000?<br>(Answer to the nearest tenth of a year) | 2 |

- (b) A total of \$15 600 is to be shared among three people A, B and C. If A is to get the smallest share of \$1600, find the value of the remaining shares when

- |      |                                                          |   |
|------|----------------------------------------------------------|---|
| (i)  | the values of the three shares form an arithmetic series | 2 |
| (ii) | the values of the three shares form a geometric series   | 2 |

- (c) Two cards are chosen at random, without replacement from the seven cards below.

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 2 | 2 | 4 | 6 | 6 | 6 | 6 |
|---|---|---|---|---|---|---|

What is the probability that

- |      |                                                                    |   |
|------|--------------------------------------------------------------------|---|
| (i)  | both cards are 2?                                                  | 1 |
| (ii) | the sum of the two numbers on the cards chosen, is greater than 8? | 2 |

## START A NEW BOOKLET

### Question 8: (12 Marks)

Marks

- (a) Two artillery guns are situated 3 kilometres apart at positions  $X$  and  $Y$  respectively. They are both aiming at a target  $T$ . 3

The angles  $XYT$  and  $YXT$  are respectively  $72^\circ$  and  $78^\circ$ .

Find the distance between the target and the gun nearer to it

- (b) Given that  $x = -\frac{1}{2}$  is one root of the following quadratic equation 3

$$mx^2 - 20x + m = 0,$$

find the exact value of the other root.

- (c) Consider the function  $y = x^3 + 3x^2 - 9x$

- (i) Find the coordinates of the stationary points. 2
- (ii) Find the domain in which the curve is concave upwards. 1
- (iii) Sketch the curve for  $-5 \leq x \leq 3$ . 3

## START A NEW BOOKLET

### Question 9: (12 Marks)

Marks

(a) Solve  $|x + 1| = 2x + 7$

3

- (b) Comment on the following reasoning ie JUSTIFY whether the following statement is TRUE or FALSE

2

“When two coins are tossed, they can either fall as two heads or two tails or as a tail and a head.

As there are three possibilities the probability of 2 heads is  $\frac{1}{3}$ .”

- (c) The acceleration  $a$  metres per second per second of a moving object is given at time  $t$  seconds ( $t \geq 0$ ) by

$$a = 2\pi^2 \cos \pi t$$

At time  $t = 0$ , the object is at the point  $x = 0$ , and travelling with velocity  $v = \pi$  metres per second.

- (i) Find the velocity  $v$  and the displacement  $x$  as a function of  $t$ , for  $t \geq 0$
- (ii) Find, for  $t$  in the range  $0 \leq t \leq 4$ , the values of  $t$  for which the object is stationary.
- (iii) Show that, for  $t$  in the range  $2 \leq t \leq 4$ , the largest value of  $x$  is

3

2

2

$$2 + \sqrt{3} + \frac{19\pi}{6}.$$

**START A NEW BOOKLET**

**Question 10: (12 Marks)**

**Marks**

(a)  $C(n) = 1000(0.8)^n + 10\,000[1 + 0.8 + (0.8)^2 + \dots + (0.8)^{n-1}]$

**3**

Find the limiting sum of  $C$  as the number of terms increases indefinitely

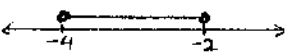
- (b) A scientist has found that the amount,  $Q(t)$ , of a substance present in a mineral at time  $t \geq 0$  satisfies

$$4 \frac{d^2 Q}{dt^2} + 4 \frac{dQ}{dt} + Q = 0$$

- (i) Verify that  $Q(t) = A(1+t)e^{-0.5t}$  satisfies this equation for any constant  $A > 0$  **4**
- (ii) If  $Q(0) = 10$  mg, find the maximum value of  $Q(t)$  and the time at which this occurs. **4**
- (iii) Describe what happens to  $Q(t)$  as  $t \rightarrow \infty$  **1**

**THIS IS THE END OF THE PAPER**

SBHS MATHEMATICS TRIAL 2002

|     |                                                                                                                                                              |      |                                                                                                                                                                                     |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ①   |                                                                                                                                                              | 2(a) |                                                                                                                                                                                     |
| (a) | $4x = 15$<br>$x = 1\frac{3}{4}, 3\frac{3}{4}$                                                                                                                | 2    | $(4, -1) (7, 3)$<br>$m = \frac{4}{3}$<br>$y+1 = \frac{4}{3}(x-4)$<br>$3y+3 = 4x-16$<br>$4x-3y-19 = 0$ as req'd                                                                      |
| (b) | 2.94                                                                                                                                                         | 2    |                                                                                                                                                                                     |
| (c) | $c(d-1) - 4(c-d)$<br>$(c-4)(d-1)$                                                                                                                            | 2    | (b) $BC(7,3)(8,1)$ $AD(4,-1)(-1,4)$<br>$m_1 = -\frac{2}{1} = -2$ $m_2 = \frac{5-4}{-1-4} = -\frac{1}{5}$<br>$m_1 \neq m_2$ $BC \nparallel AD$                                       |
| (d) | $4k+1 = 29$<br>$k = 7$                                                                                                                                       | 2    | (c) $DC(1,9) (7,3)$<br>$m = -\frac{6}{6} = -1$<br>$\text{grad } AC = \frac{4}{3} \rightarrow \text{from (a)}$<br>$m_1 m_2 = -1$ $DC \perp AC$<br>$\therefore \angle ACD = 90^\circ$ |
| (e) | $x+3 \leq 1$ or $-x-3 \leq 1$<br>$x \leq -4$ $-x \leq 4$<br>$x \geq -4$<br> | 2    | (d) $AC(4,-1) (7,3)$<br>$d = \frac{ a(x_1) + b(y_1) + c }{\sqrt{a^2+b^2}}$<br>$= \frac{ 4(-1) - 3(-1) - 19 }{\sqrt{16+9}}$<br>$= \frac{10}{5} = 2$                                  |
| (f) | $\frac{1}{\sqrt{3}+2} \times \frac{\sqrt{3}-2}{\sqrt{3}-2}$<br>$= \frac{\sqrt{3}-2}{-1}$<br>$-\sqrt{3}+2 = a\sqrt{3}+b$<br>$a = -1$ $b = 2$                  | 2    | (e) $d = \frac{ ax_1 + by_1 + c }{\sqrt{a^2+b^2}}$<br>$4x-3y-19=0$ (8,1)<br>$d = \frac{ 4(8) - 3(1) - 19 }{\sqrt{16+9}}$<br>$= \frac{10}{5} = 2$                                    |
|     |                                                                                                                                                              | (f)  | length $DC(-1,9) (7,3)$<br>$d = \sqrt{8^2+6^2} = 10$<br>Area = $\triangle DCA + \triangle ACB$<br>$A = \frac{1}{2} \times 10 \times 5 + \frac{1}{2} \times 5 \times 2$<br>$= 30$    |

# Mathematics Trial, Assessment Task #4, 2002

1 3. (a) i.  $7 \times 2(2x-1)^6 = 14(2x-1)^6$ .

2 ii.  $\frac{5}{4+5x}$ .

2 iii.  $2x \sin x + x^2 \cos x$ .

1 (b)  $\int \sec^2 3x dx = \frac{1}{3} \tan 3x + c$ .

3 (c)  $\int_1^{e^2} \frac{3}{x} dx = 3 [\ln x]_1^{e^2}$   
 $= 3 \{2 - 0\}$   
 $= 6$ .

3 (d) Area  $= \int_0^{\frac{1}{2}} e^{2x} dx$   
 $= \left[ \frac{e^{2x}}{2} \right]_0^{\frac{1}{2}}$   
 $= \frac{1}{2} \{e^2 - 1\}$ .

2 4. (a) i.  $x^2 - 2x + 1 = 8y - 25 + 1$ ,  
 $\therefore (x-1)^2 = 4 \times 2(y-3)$ .

1 ii. (1,3).

1 iii. (1,5).

1 iv.  $y = 1$ .

4 (b) First method: using  $\lambda$ ,  
 $x + 2y - 6 + \lambda(x - y) = 0$ ,  
 $(2 - \lambda)y = -(1 + \lambda)x + 6$ .  
 The slope of line  $3x - 2y + 7 = 0$  is  $\frac{3}{2}$ .  
 $\therefore \frac{3}{2} = \frac{\lambda + 1}{\lambda - 2}$ ,  
 $3\lambda - 6 = 2\lambda + 2$ ,  
 $\lambda = 8$ .  
 $\therefore$  New line is  $x + 2y - 6 + 8x - 8y = 0$ .  
 i.e.  $9x - 6y - 6 = 0$ ,  
 $3x - 2y - 2 = 0$ .

Second method: first find the intersection,  
 $-y = 2y - 6$ ,  
 $y = 2$ ,  
 and  $x = 2$ .

$\therefore$  Intersection is (2,2).  
 The slope of line  $3x - 2y + 7 = 0$  is  $\frac{3}{2}$ .  
 $\therefore$  New line is:  $y - 2 = \frac{3}{2}(x - 2)$ ,  
 $2y - 4 = 3x - 6$ ,  
 i.e.  $3x - 2y - 2 = 0$ .

3 (c) Area  $\triangle XYZ = \frac{1}{2} \times 4 \times 4$ ,  
 $= 8 \text{ unit}^2$ .  
 Area of Sector  $XYW = \frac{1}{2} \times 4^2 \times \frac{\pi}{4}$ ,  
 $= 2\pi \text{ unit}^2$ .  
 $\therefore$  Shaded area  $= 8 - 2\pi \text{ unit}^2$ .

## TRIAL 2U HSC 2002 SOLUTIONS

### QUESTION 5

a)  $y = x^2 - 3x + 6$

(2 marks)  $y' = 2x - 3$

at (3,5)

$y' = 2 \times 3 - 3$

$= 3$

hence  $m_2 = \frac{-1}{m_1}$   
 $= \frac{-1}{3}$

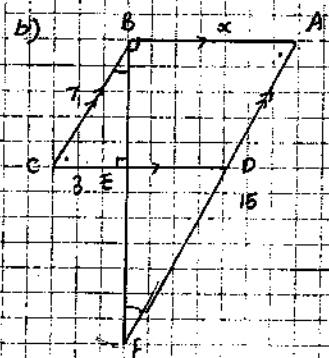
using

$y - y_1 = m(x - x_1)$

$y - 5 = \frac{-1}{3}(x - 3)$

$3y - 15 = -x + 3$

$\therefore x + 3y - 18 = 0$



i) In  $\triangle CBE$  and  $\triangle AFB$

(3 marks)  $\angle CEB = \angle AFB$  (alt.  $\angle$ s,  $AB \parallel EF$ )

$\angle BCE = \angle BAF$  (opp.  $\angle$ s,  $BC \parallel AF$ )

$\angle CBE = \angle AFB$  (sum of  $\angle$ s)

$\therefore \triangle CBE \parallel \triangle AFB$  (equiangular)

ii) In  $\triangle CBE$  and  $\triangle AFB$

(2 marks)  $\frac{CB}{CE} = \frac{AF}{AB}$  (corr. sides of  $\parallel \triangle$ s)

$\frac{7}{8} = \frac{15}{x}$

$7x = 120$

$x = 17\frac{1}{7}$

$\therefore AB = 17\frac{1}{7} \text{ cm}$

c)  $y = x^2 - 1$

(6 marks)  $y^2 = (x^2 - 1)^2$   
 $= x^4 - 2x^2 + 1$

$$V = \pi \int y^2 dx$$

$$= 2\pi \int_0^1 (x^4 - 2x^2 + 1) dx$$

$$= 2\pi \left[ \frac{x^5}{5} - \frac{2x^3}{3} + x \right]_0^1$$

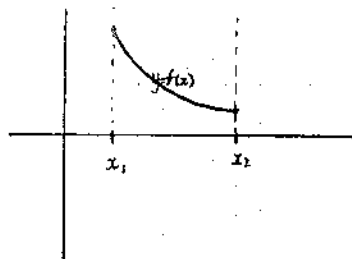
$$= 2\pi \left[ \left( \frac{1^5}{5} - \frac{2(1)^3}{3} + 1 \right) - \left( \frac{0^5}{5} - \frac{2(0^3)}{3} + 0 \right) \right]$$

$$= 2\pi \left( \frac{1}{5} - \frac{2}{3} + 1 \right)$$

$$= 2\pi \left( \frac{8}{15} \right)$$

$$V = \frac{16\pi}{15} \text{ units}^3$$

d)  $y = f(x)$   $x_1 \leq x \leq x_2$   
 (2 marks)  $f(x) > 0$ ,  $f'(x) < 0$ ,  $f''(x) > 0$



$f(x)$  above  $x$ -axis  
 $f'(x)$  negative gradient  
 $f''(x)$  concave up

## QUESTION 6

a)  $y = x^3 + mx + n$   $P(1,6)$  when  $y' = 0$

(3 marks)

$$y' = 3x^2 + m$$

at  $P(1,5)$

$$0 = 3(1)^2 + m$$

$$\therefore m = -3$$

$$y = x^3 - 3x + n$$

at  $P(1,5)$

$$5 = (1)^3 - 3(1) + n$$

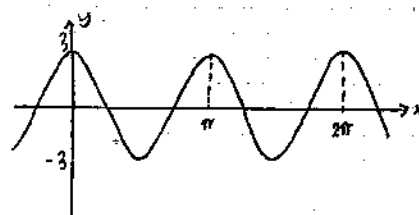
$$5 = 1 - 3 + n$$

$$7 = n$$

$$\therefore n = 7$$

b)

(2 marks)



$$y = a \cos mx$$

where amplitude  $(a) = 3$

$$\therefore y = 3 \cos 2x$$

period  $T = \frac{2\pi}{m}$

$$\pi = \frac{2\pi}{m}$$

$$\therefore m = 2$$

c)

(2 marks)

$$\ln 2 + 2 \ln 18 - \frac{3}{2} \ln 36$$

$$= \ln 2 + \ln 324 - \ln 216$$

$$= \ln \left( \frac{2 \times 324}{216} \right)$$

$$= \ln 3$$

d)

(marks)

$$\sqrt{\frac{1 - \cos^2 \theta}{1 + \tan^2 \theta}}$$

Using  $\cos^2 \theta + \sin^2 \theta = 1$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$1 + \tan^2 \theta = \sec^2 \theta = \frac{1}{\cos^2 \theta}$$

$$= \sqrt{\frac{\sin^2 \theta}{\frac{1}{\cos^2 \theta}}}$$

$$= \sqrt{\sin^2 \theta \cos^2 \theta}$$

$$= \sin \theta \cos \theta$$

e)

(3marks)

$$\int_{-1}^7 f(x) dx \approx \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_5) + 2(y_2 + y_4)]$$

where  $h = \frac{b-a}{n}$

$$= \frac{7 - (-1)}{4}$$

$$= 2$$

$$y_0 = 5$$

$$y_1 = 9$$

$$y_2 = 2$$

$$y_3 = -1$$

$$y_4 = -6$$

$$\int_{-1}^7 f(x) dx \approx \frac{2}{3} [(5 - 6) + 4(9 - 1) + 2(2)]$$

$$\approx \frac{2}{3} (-1 + 32 + 4)$$

$$\approx \frac{70}{3}$$

$$\approx 23\frac{1}{3}$$

## QUESTION 7

d) i)

$$V = 50000 e^{-0.25t}$$

(1mark)

when  $t=0$ ,  $V=?$

$$V = 50000 e^{-0.25 \times 0}$$

$$= 50000$$

$\therefore$  value of new car is \$50000

ii)

when  $t=3$ ,  $V=?$

(2mark)

$$V = 50000 e^{-0.25 \times 3}$$

$$= 23618.33$$

$\therefore$  value of car after 3 years is \$23600 (nearest \$100)

iii)

$V = 15000$   $t=?$

(2marks)

$$15000 = 50000 e^{-0.25t}$$

$$\frac{15000}{50000} = e^{-0.25t}$$

$$0.3 = e^{-0.25t}$$

$$\ln 0.3 = -0.25t$$

$$t = \frac{\ln 0.3}{-0.25}$$

$$= 4.815$$

$\therefore$  time taken for the car to be worth \$15000 is 4.8 years (nearest tenth of a year).

b) i)  
(2 marks)  
 $S_n = a + (a+d) + (a+2d)$

where  $S_n = 15600$

$a = 1600$

$5600 = 1600 + (1600+d) + (1600+2d)$

$10800 = 3d$

$d = 3600$

$A = \$1600$     $B = \$5200$     $C = \$8800$

ii)  $S_n = \frac{a(r^n-1)}{r-1}$   
(2 marks)

$S_n = a + ar + ar^2$

$15600 = 1600 + 1600r + 1600r^2$

$14000 = 1600r^2 + 1600r$

$0 = 8r^2 + 8r - 70$

$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{-8 \pm \sqrt{64 + 2240}}{16}$

$\therefore r = \pm 2\frac{1}{2}$

$r = 2.5$

$\therefore A = \$1600$     $B = \$4000$     $C = \$10000$

c) i)  $P(2,2) = \frac{2}{7} \times \frac{1}{6}$   
(1 mark)  
 $= \frac{1}{21}$

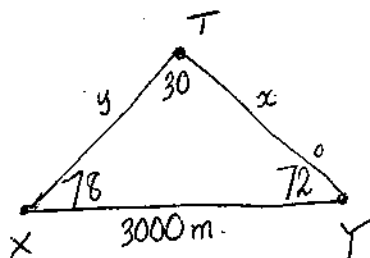
ii)  $P(1+6) = \frac{1}{7} \times \frac{4}{6}$   
(2 marks)  
 $= \frac{4}{42}$

$P(6+6) = \frac{4}{7} \times \frac{3}{6}$   
 $= \frac{12}{42}$

$P(6+4) = \frac{4}{7} \times \frac{1}{6}$   
 $= \frac{4}{42}$

$P(1+8) = \frac{8}{42} + \frac{12}{42}$   
 $= \frac{20}{21}$

(a)



$$\frac{y}{\sin 72^\circ} = \frac{3000}{\sin 30^\circ}$$

$$y = \frac{3000 \times \sin 72^\circ}{\sin 30^\circ}$$

$$\therefore 5706 \text{ m} \div 5.706 \text{ km}$$

(3)

$$(b) mx^2 - 20x + m = 0$$

$$\text{let } x = -\frac{1}{2} \quad \frac{m}{4} + 10 + m = 0$$

$$m + 40 + 4m = 0$$

$$5m = -40$$

$$m = -8$$

$$\text{So we have } -8x^2 - 20x - 8 = 0$$

$$\text{and } -\frac{1}{2} + \alpha = -\frac{20}{-8}$$

$$r(1x+1)(x+2)=0$$

$$x = -\frac{1}{2}, x = -2$$

$$\alpha = -\frac{20}{8} + \frac{1}{2}$$

$$= -2$$

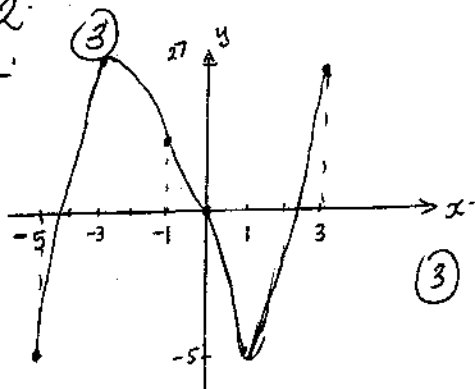
Other root is -2.

$$(c) y = x^3 + 3x^2 - 9x$$

$$y' = 3x^2 + 6x - 9$$

$$= 3(x^2 + 2x - 3)$$

$$y'' = 6x + 6 = 6(x+1)$$



$$3(x^2 + 2x - 3) = 0$$

$$x^2 + 2x - 3 = 0$$

$$(x+3)(x-1) = 0$$

$$x = -3, x = 1$$

$$y = 27, y = -5$$

stat points are  
(-3, 27) and (1, -5)

(2)

(ii) Curve is concave  
upwards when  $y'' > 0$

$$6(x+1) > 0$$

$$x+1 > 0$$

$$x > -1 \quad (1)$$

$$(iii) (-3, 27) y'' < 0 \text{ max}$$

$$(1, -5) y'' > 0 \text{ min}$$

(-1, 1) inflection

$$\text{At } x = -5, y = -5$$

$$x = 3, y = 27$$

$$(a) |x+1| = 2x+7$$

$$x+1 = 2x+7$$

$$-6 = x$$

LHS  $\neq$  RHS

$$x+1 = -2x-7$$

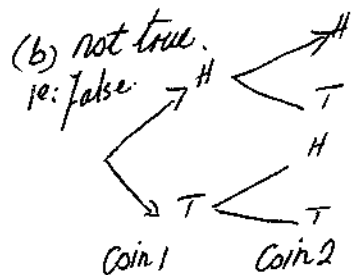
$$3x = -8$$

$$x = -\frac{8}{3}$$

$$\text{LHS} = 1\frac{2}{3}$$

$$\text{RHS} = 1\frac{2}{3}$$

$$x = -\frac{8}{3} = -2\frac{2}{3} \quad (-2.6) \quad (3)$$



4 possibilities  
not 3. (2)

$$(c) (i) a = 2\pi^2 \cos \pi t$$

$$\text{data } t=0$$

$$x=0$$

$$V=\pi$$

$$V = \int 2\pi^2 \cos \pi t \, dt$$

$$= \frac{2\pi^2}{\pi} \int \pi \cos \pi t \, dt$$

$$V = 2\pi \sin \pi t + C_1$$

$$\pi = C_1$$

$$V = 2\pi \sin \pi t + \pi$$

$$x = \int (2\pi \sin \pi t + \pi) \, dt$$

$$x = -2 \cos \pi t + \pi t + C_2$$

(ii) when  $V=0$

$$2\pi \sin \pi t + \pi = 0$$

$$2\pi \sin \pi t = -\pi$$

$$\sin \pi t = -\frac{\pi}{2\pi} = -\frac{1}{2}$$

$$\sin \pi t = -\frac{1}{2}$$

$$\pi t = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$t = \frac{7\pi}{6\pi} = \frac{7}{6} \text{ sec}$$

$$t = \frac{11\pi}{6\pi} = \frac{11}{6}$$

$$\text{also } t = \frac{19}{6}, \frac{23}{6}$$

$$(iii) t = \frac{19}{6} \quad x = -2 \cos \frac{19\pi}{6}$$

$$= -2 \cos \frac{7\pi}{6}$$

$$= 2 + \sqrt{3} + 19$$

$$t = \frac{23}{6} \quad x = -2 \cos \frac{23\pi}{6}$$

$$= -2 \cos \frac{11\pi}{6}$$

$$= -2 \times \frac{\sqrt{3}}{2} + \frac{23\pi}{6}$$

$$= 2 + \sqrt{3} + 19$$

$$t = \frac{23}{6} \quad x = -2 \cos \frac{23\pi}{6}$$

$$= -2 \cos \frac{11\pi}{6}$$

$$= -2 \times \frac{\sqrt{3}}{2} + \frac{23\pi}{6}$$

$$= 2 + \sqrt{3} + 19$$

$$t = \frac{23}{6} \quad x = -2 \cos \frac{23\pi}{6}$$

$$= -2 \cos \frac{11\pi}{6}$$

$$= -2 \times \frac{\sqrt{3}}{2} + \frac{23\pi}{6}$$

$$= 2 + \sqrt{3} + 19$$

$$\text{At } t = \frac{19}{6} \quad x \text{ is large}$$

$$\text{so } x = -2 \cos \pi t + \pi t$$

$$② C(n) = 1000(0.8)^n + 10,000[1 + .8 + .8^2 + \dots + .8^n]$$

$$\text{as } n \rightarrow \infty \quad 1000(0.8)^n \rightarrow 0$$

$$\text{now } 10,000[1 + .8 + .8^2 + \dots + .8^{n-1}]$$

$$\left. \begin{array}{l} \text{GP } a=1 \\ r=.8 \\ n=n \end{array} \right\} S_n = \frac{1(.8^n - 1)}{.8 - 1} = \frac{.8^n - 1}{-.2} = \frac{1 - .8^n}{.2}$$

$$\text{so we have } 10,000 \times 5(1 - .8^n) = 50,000(1 - .8^n) \quad (3)$$

$$\text{as } n \rightarrow \infty \quad 50,000(1 - .8^n) \rightarrow 50,000$$

$$⑥ (i) \phi(t) = A(1+t)e^{-0.5t}$$

$$\phi(t) = Ae^{-0.5t} + Ate^{-0.5t}$$

$$\begin{aligned} \phi'(t) &= -0.5Ae^{-0.5t} + At \times -0.5e^{-0.5t} + Ae^{-0.5t} \\ &= 0.5Ae^{-0.5t} - 0.5Ate^{-0.5t} \end{aligned}$$

$$\begin{aligned} \phi''(t) &= -0.25Ae^{-0.5t} = (0.5At \times -0.5e^{-0.5t} + 0.5Ae^{-0.5t}) \\ &= -0.25Ae^{-0.5t} + 0.25Ate^{-0.5t} - 0.5Ae^{-0.5t} \\ &= -0.75Ae^{-0.5t} + 0.25Ate^{-0.5t} \end{aligned}$$

$$\begin{aligned} &4(-0.75Ae^{-0.5t} + 0.25Ate^{-0.5t}) + 4(0.5Ae^{-0.5t} - 0.5Ate^{-0.5t}) + Ae^{-0.5t} + Ate^{-0.5t} \\ &= 0 \quad (4) \end{aligned}$$

$$v) (ii) \psi(0) = 10$$

$$\text{So when } t=0 \quad A(1+t)e^{-0.5t} = 10$$

$$A(1+0)e^{-0.5 \times 0} = 10$$

$$A = 10$$

$$\text{So } \phi(t) = 10(1+t)e^{-0.5t}$$

$$\begin{aligned} \phi'(t) &= 0.5 \times 10e^{-0.5t} - 0.5 \times 10te^{-0.5t} \\ &= 5e^{-0.5t} - 5te^{-0.5t} \end{aligned}$$

$$= 5e^{-0.5t}(1-t)$$

$$\text{let } \phi'(t) = 0 \Rightarrow t = 1$$

$$\begin{aligned} \text{Sub } t=1, \quad \phi''(t) &= -0.75 \times 10e^{-0.5} + 0.25 \times 10 \times 1e^{-0.5} \\ &= -7.5e^{-0.5} + 2.5e^{-0.5} \\ &= -5e^{-0.5} < 0 \end{aligned}$$

At  $t=1$ ,  $\phi(t)$  is a max.

$$\begin{aligned} \text{When } t=1, \quad \phi(t) &= 10(2)e^{-0.5} \\ &= \frac{20}{e^{0.5}} \approx 12.13 \quad (4) \end{aligned}$$

$$\begin{aligned} (iii) \phi(t) &= 10(1+t)e^{-0.5t} \\ &= 10e^{-0.5t} + 10te^{-0.5t} \\ &= \frac{10}{e^{0.5t}} + \frac{10t}{e^{0.5t}} \quad \text{As } t \rightarrow \infty \quad \phi(t) \rightarrow 0 \quad (i) \end{aligned}$$